

1. Project Introduction

Project Title:

Load Testing of BlazeDemo Flight Booking Application Using Gatling

1.1 Overview

With the rapid growth of web-based applications, ensuring application performance and reliability has become a critical aspect of software quality assurance. Modern applications are expected to provide fast response times, high availability, and consistent user experience even when accessed by a large number of users simultaneously. Performance issues such as slow response times, system crashes, server overload, and transaction failures can negatively impact user satisfaction and business operations.

To prevent such issues, performance testing is conducted to evaluate how an application behaves under different workload conditions. Performance testing helps identify bottlenecks, measure system responsiveness, validate scalability, and ensure that the application can handle expected user traffic.

This project focuses on performing load testing of the BlazeDemo Flight Booking Application using Gatling. The project was designed to simulate real user activities and measure the application's performance under load conditions. The testing process involved creating multiple test scenarios, executing virtual user traffic, and analyzing detailed performance reports generated by Gatling.

1.2 About BlazeDemo

BlazeDemo is a sample web-based flight booking application widely used for testing and automation practice. The application simulates a simple airline reservation system and provides a complete booking workflow that allows users to search available flights, select flights, and complete the booking process.

Official Website:

<https://blazedemo.com>

The application contains multiple web pages connected through a business workflow. Users can select departure and destination cities, search available flights, choose a preferred flight, provide passenger details, and confirm the booking.

Because of its simple design and realistic booking workflow, BlazeDemo is an excellent application for demonstrating functional testing, automation testing, API testing, and performance testing concepts.

1.3 Importance of Load Testing

Load testing is a type of performance testing that evaluates application behavior under expected user load conditions. The primary objective is to determine whether the application can maintain acceptable performance levels when multiple users access the system simultaneously.

Load testing provides answers to important questions such as:

- How fast does the application respond under load?
- How many users can the system handle simultaneously?
- Does the application remain stable during heavy traffic?
- Are there any performance bottlenecks?
- Does response time increase significantly as load increases?

By identifying such issues early, organizations can improve system performance before releasing applications to production environments.

1.4 About Gatling

Gatling is an open-source performance testing tool designed for load testing web applications, APIs, and enterprise systems. It is widely used because of its ability to simulate large numbers of virtual users while consuming relatively fewer system resources.

Key features of Gatling include:

- High-performance load generation
- Detailed HTML reporting
- Real-time metrics collection
- Java, Scala, and Kotlin support
- Easy scenario creation
- Scalability and flexibility

In this project, Gatling Java DSL was used to create and execute load testing scenarios for the BlazeDemo application.

1.5 Project Scope

The scope of this project is limited to evaluating the performance of the BlazeDemo Flight Booking Application using simulated user traffic.

The project covers the following user workflows:

Scenario 1 – Search Flights

This scenario validates the flight search functionality of the application.

Workflow:

Home Page

↓

Select Departure City

↓

Select Destination City

↓

Find Flights

The objective of this scenario is to measure the application's response time while retrieving available flights.

Scenario 2 – Choose Flight

This scenario validates the flight selection process.

Workflow:

Home Page

↓

Find Flights

↓

Choose Flight

The objective of this scenario is to evaluate how efficiently the application processes flight selection requests under load.

Scenario 3 – Complete Booking

This scenario validates the complete end-to-end booking process.

Workflow:

Home Page

↓

Find Flights

↓

Choose Flight

↓

Enter Passenger Details

↓

Purchase Flight

↓

Confirmation Page

This scenario represents a real business transaction and is considered the most important workflow in the application.

1.6 Project Goal

The main goal of this project is to perform advanced load testing of the BlazeDemo Flight Booking Application using Gatling and prepare a detailed performance report.

The project aims to:

- Simulate concurrent users
- Execute multiple business workflows
- Measure response times
- Evaluate throughput
- Monitor request success rates
- Identify performance bottlenecks
- Generate detailed performance reports
- Assess application stability under load

The results obtained from the execution of the test scenarios provide valuable insights into the performance characteristics of the application and help determine its ability to handle user traffic efficiently.